

AUDIO / VIDEO TECH VL04





ANALOG

resistors in electrical circuits
(potentiometers, faders)

VCA (voltage controlled
amplifier), analog mixing desks,
modular synths

CONTROL OF PARAMETERS
analog vs digital



DIGITAL

automation in DAWs

MIDI

OSC

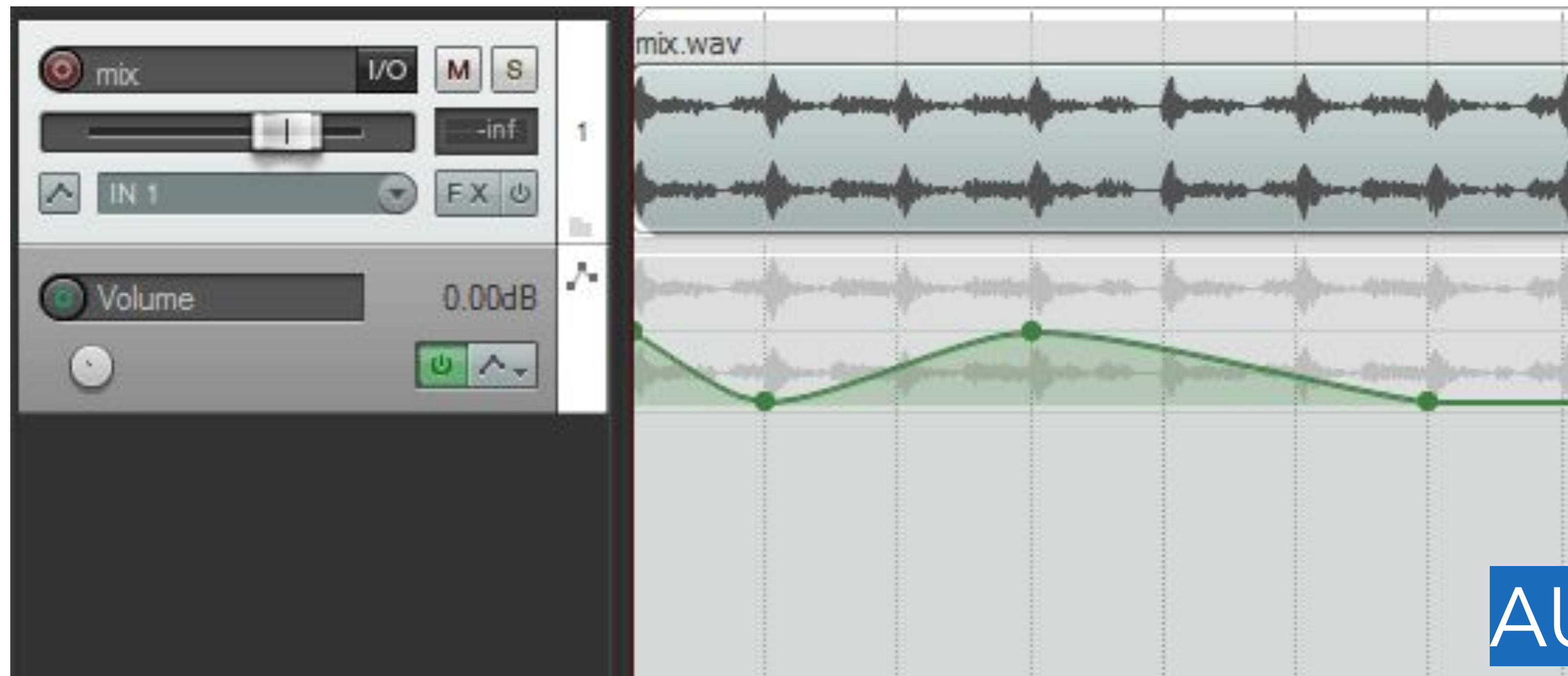
web interfaces

DCA (digitally controlled
amplifier)

DMX

CONTROL OF PARAMETERS
analog vs digital

- automation is DAW specific
- volume, pan, sends and almost all effect parameters available
- envelopes can be recorded and/or edited
- control through mouse, LFOs, MIDI, OSC, audio from another track (sidechain)



AUTOMATION

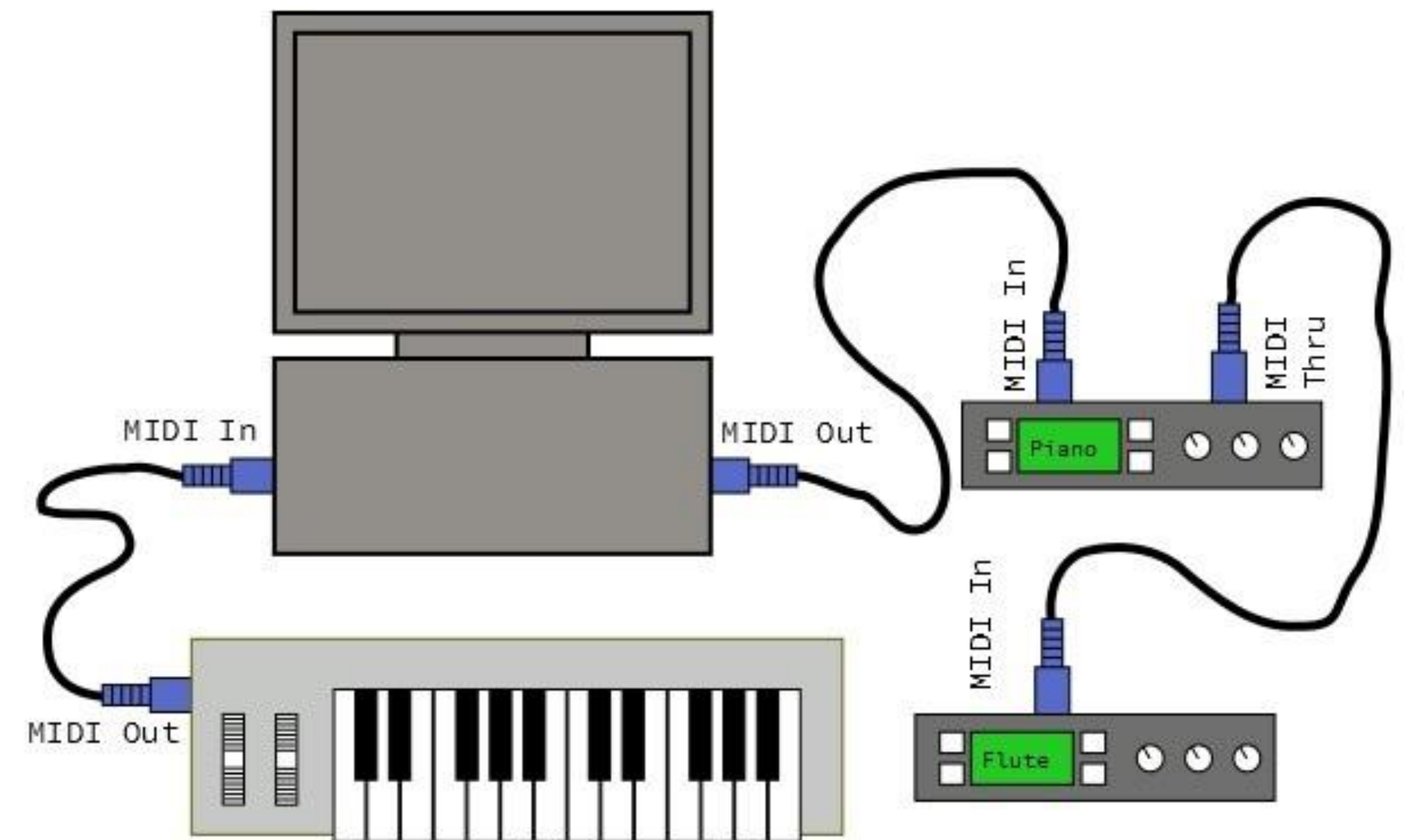
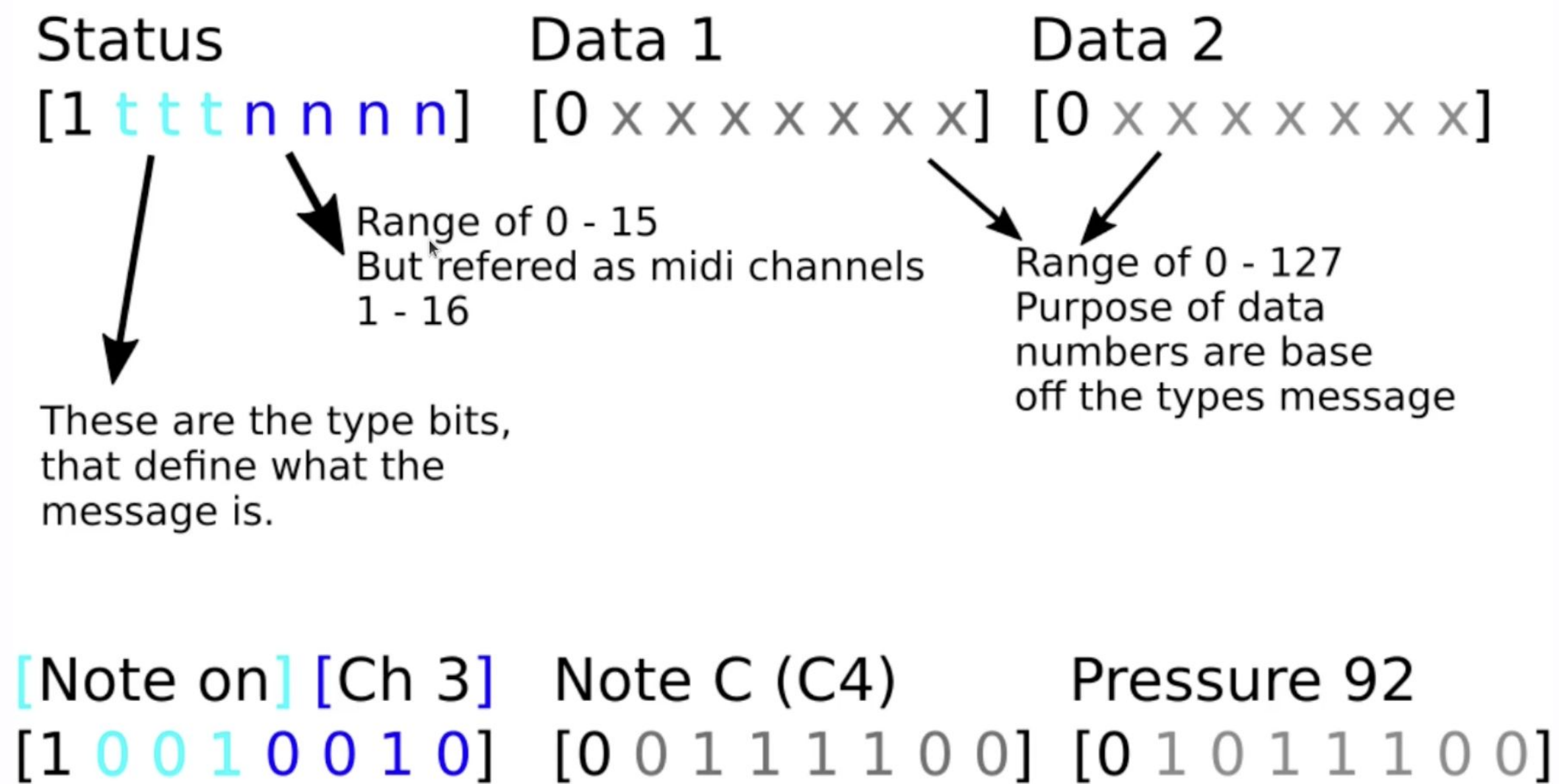
- Musical Instrument Digital Interface
- serial, asynchronous, unidirectional protocol
- invented in 1982 for using one keyboard for different sound modules (for layering or splitting)
- can be used to control pitch (note number), velocity and length (note on and note off) of samplers and synthesizers
- control parameters (cc) for musical expression
- midi files (.mid, .midi, .smf, .rmi) can be opened in notation software (eg. MuseScore)
- MIDI clock (MC) und MIDI time code (MTC) available for synching multiple devices



5-DIN MIDI
In/Out



- 1 status byte followed by 2 data bytes
- 16 channels
- resolution limited to 128 steps



MIDI

- cc (control change / continuous controllers) for musical expression
controller number + value (0-127)
- CC 1: Modulation Wheel
- CC 7: Main Volume
- CC 10: Pan
- CC 11: Expression
- CC 64: Sustain Pedal
- pc (program change) for switching to another preset



MIDI

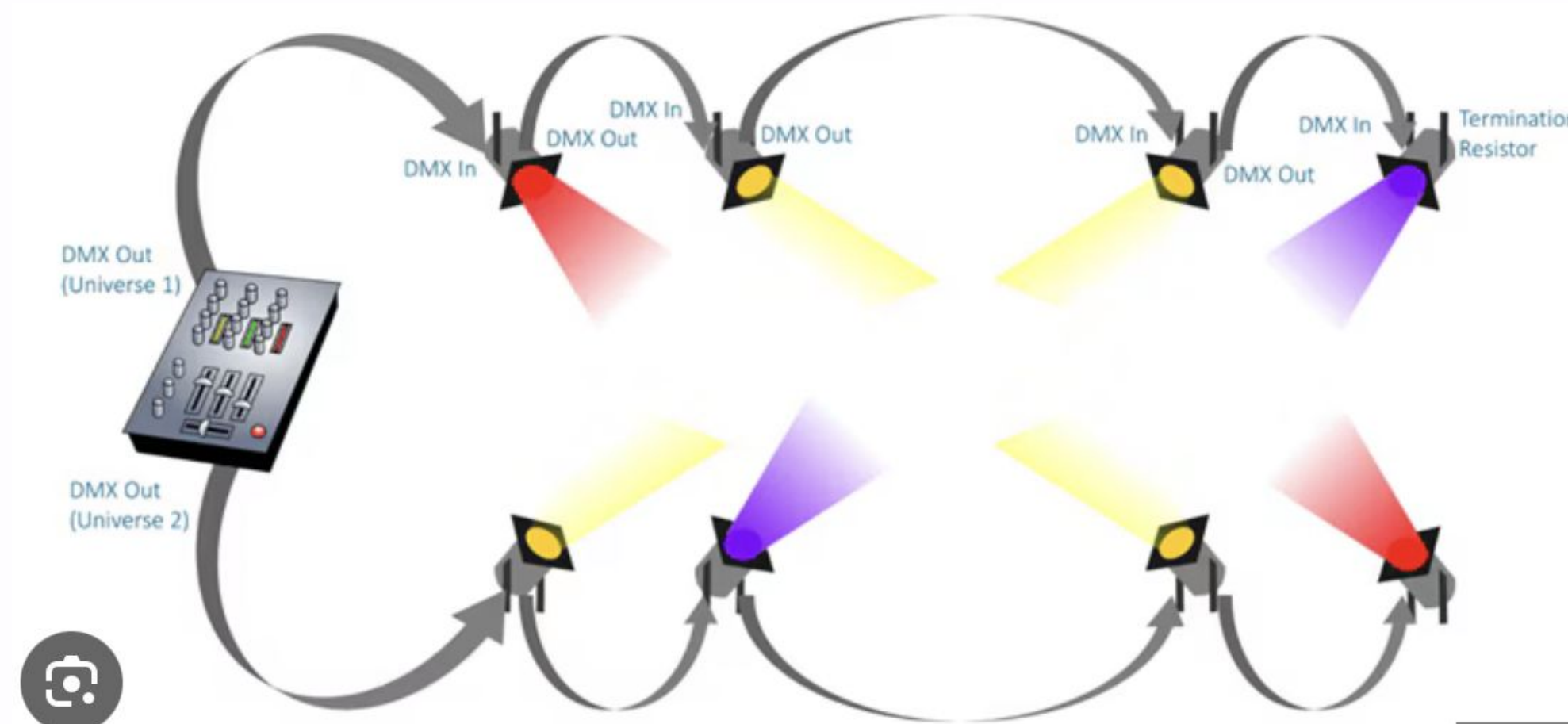
Open Sound Control

- more flexible alternative to MIDI, higher resolution
- protocol for communication among computers, sound synthesizers, and other multimedia devices that is optimized for modern networking technology
- <https://opensoundcontrol.stanford.edu/>



DMX / DMX512

- digital multiplex, asynchronous, serial protocol
- for controlling stage lighting and effects
- 1 controller, devices can be daisy chained (each with in & out)
- 1 universe = 512 channels
- values from 0 - 255 (8bit)
- 2 channels (16bit) for moving lights
- even more channels for more parameters and effects



DMX



- scenes can be triggered by a GO button
- they can contain sound, video, DMX, fades, etc.
- scenes run in parallel until they end or are being stopped by another cue
- alternatives: ShowCueSystem (PC), Audio Cues (Android)



- objects based DAW for DJs with workarounds for theatre sound
- scenes are triggered (fired) through master section
- fades on cue possible through automation written into empty objects on separate control tracks
- complex arrangements can be modified on cue
- alternative: Playtime/ReaLearn for Reaper

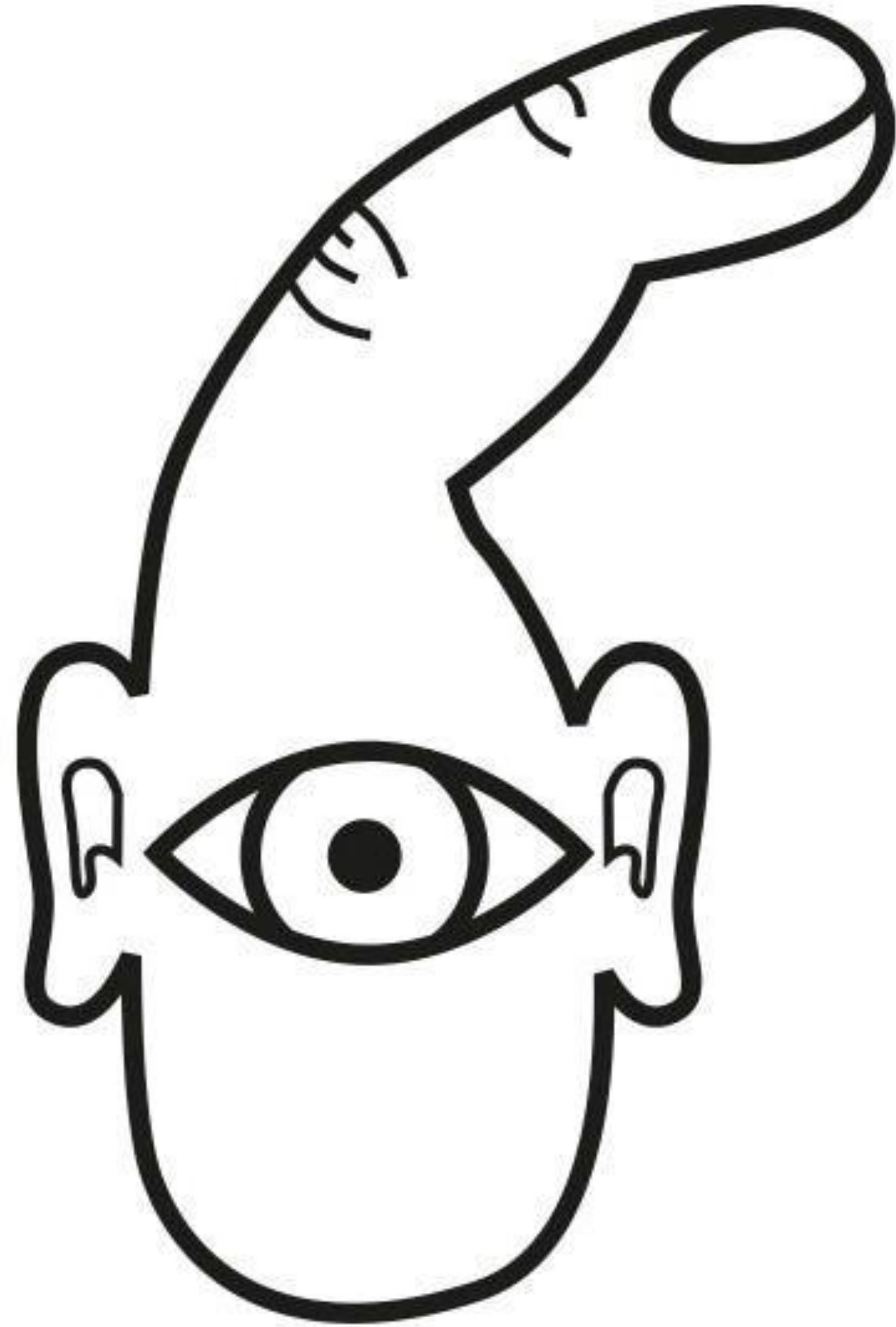
SOFTWARE SOLUTIONS
Ableton Live



- highly customizable control surface for all platforms
- alternative: OpenStageControl

SOFTWARE SOLUTIONS
TouchOSC

How the computer sees us...





ALTERNATIVE MUSICAL
INSTRUMENT INTERFACES